

First Demonstration of Monolithic 3-Tier Nanosheet Transistor Stacking with Split Gate Featuring Tri-State Inverter/Half SRAM Functionalities

Bo-Wei Huang¹, Ying-Qi Liu², Ching-Wang Yao², Wei-Jen Chen¹, Min-Kuan Lin², Xin-Yuan Lin², Chun-Yi Cheng¹, Yi Huang³, Ding-Wei Lin¹, Chih-Hsuan Lu¹, Tsung-Han Tsai³, and C. W. Liu^{1,2,3*}

¹Graduate Institute of Electronics Engineering, ²Graduate School of Advanced Technology,

and ³Graduate Institute of Photonics and Optoelectronics, National Taiwan University, Taipei, Taiwan. *E-mail: cliu@ntu.edu.tw

Abstract

Monolithic three-tier transistor stackings with GeSi nanosheets, split gate, and multiple pn layer isolation are demonstrated for the first time. The split-gate process enables one common-gate CFET and one nFET underneath to achieve 6T STAM design with 3T footprint. The functionalities of tri-state inverters and half-SRAM (1PD/1PU/1PG) of monolithic three-tier transistor stacking are successfully demonstrated.

Introduction

3D transistor stacking, including CFET, has been reported as an essential option for extending the technology scaling [1-12]. Several approaches, including monolithic [1-9] and sequential [10-15], have been explored to investigate the potential. The split-gate CFETs are necessary to offer the design flexibility [9, 16]. Stacking more than two tiers can reduce the footprint but requires a novel cell design. Sequential processes have been reported to stack transistors more than two tiers [11, 12] by repeating bonding processes [13-15]. Although the monolithic approach has the advantage of low cost due to the self-alignment process [17], the process complexity makes it challenging to have three-tier or more stacking.

In this work, monolithic three-tier transistor stacking is fabricated by co-optimizing epitaxy, etching, and the split-gate processes [9]. With three-tier transistor stacking and the split-gate process, half SRAM and tri-state inverter circuits can be realized using external wiring connectivity. To electrically isolate the stacked FETs in the source/drain (S/D), multiple pn layer isolation was invented by epitaxial growth to replace the dielectric isolation for simplicity [1-3, 9]. The heavily doping sacrificial layers (SLs) provide the doping in S/D region for low S/D resistance without S/D regrowth.

Epilayer Design and Device Fabrication

After the SOI was thinned down to 20nm, the undoped Ge buffer was grown with *in-situ* annealing for the high-quality epi channels. The 18nm Ge_{0.76}Si_{0.24} channel layer sandwiched by P-doped SLs were grown for the bottom FET. The 78nm Ge:P and 130nm Ge:B SLs were grown thrice. This pn layer isolation can simplify the process without the complex integration of dielectric layers for S/D isolation [1-3]. The Ge_{0.76}Si_{0.24} channel layer and Ge:B SLs were grown for the middle pFET. The multiple pn layer isolation was grown again for the top and middle FETs isolation. Finally, the Ge_{0.76}Si_{0.24} channel layer and Ge:P SLs were grown for the top nFET. The abrupt interface between channel layers and SLs are observed in Fig. 2. The XRD peak of the Ge buffer shows 0.05% tensile strain (Fig. 3(a)) due to the mismatch of thermal expansion coefficients between Si and Ge. The Ge_{0.76}Si_{0.24} channels are fully tensile strained on the Ge buffer with a strain of 1.12% (Fig. 3(b)). Strong photoluminescence (PL) spectra indicate high-quality epi growth (Fig. 3(c)). The process for monolithic three-tier transistor stacking can be divided into the common-gate process and the split-gate process (Fig. 4). The SiO₂ was deposited by PECVD as hard mask after the 18-layer CVD epitaxy. The channels and S/D regions are defined by e-beam lithography. Then, the fin formation was performed by Cl₂-based RIE. Note that the fin height is as tall as ~1.6μm due to three-tier transistor stacking. The field oxide was then deposited and patterned. The channel release was then performed by the H₂SO₄+H₂O₂ wet etching to have better etching selectivity compared to HNO₃ wet etching (Fig. 5). The Hf_{0.2}Zr_{0.8}O₂ gate dielectrics were conformally deposited

around the nanosheets by PEALD, followed by PEALD TiN [18-21] and thick PVD TiN as gate metal pad. The step formation (Fig. 1) for each FETs contacts were then performed twice by RIE [1-3, 9]. The contacts were formed by Pt. The following process steps were performed for the split-gate scheme. The SiO₂ isolation between gates were deposited, with the BOE etching to reveal top and middle gate metal. The top and middle gate metal were etched by heated NH₄OH+H₂O₂ [9, 22, 23]. The TiN was then deposited by PEALD for top and middle gate to form common gate. Finally, the contacts were revealed by BOE etching.

3T footprint SRAM design and Experiment Results

One of 3T footprint SRAM designs using monolithic three-tier transistor stacking is shown in Fig. 6. 2T footprint is used for inverters and two pass gates, while 1T footprint is used for cross couple connections. The I_D-V_{GS} transfer curves of each transistor in the three-tier stacking are shown in Fig. 7. The small I_{ON}/I_{OFFS} are due to the narrow bandgap (~690meV [24]) of Ge_{0.76}Si_{0.24} and the isolation leakage of multiple pn layer isolation, where one nFET turns off and the other stacked pFET turns on simultaneously. The V_{TS} are 0.79V, -0.6V, and 0.7V. The V_{TS} are positive for nFETs and negative for the pFET because the metal work function of TiN is near the middle of the GeSi bandgap [25]. The top, middle, and bottom nFET has channel width (W_{CH}) of 90nm, 120nm, and 150nm. The tall tapered fin (~1.6μm) results in the different W_{CH} of FETs (Fig. 8) since the tall fin has long etching time at top but short time at bottom. The channels are surrounded by Hf_{0.2}Zr_{0.8}O₂ dielectric and TiN gate metal for GAA structures, as confirmed by the EDS mapping (Fig. 9).

With the external wiring, the monolithic three-tier transistor stacking can be connected as one tri-state inverter (Fig. 10). The tri-state inverter produces three output states: a logic high, a logic low, and a high impedance state. When a high voltage (V_{DD}) is applied at input A, the tri-state inverter acts as a standard inverter. On the other hand, when a low voltage is applied at input A, the output B is in a high impedance state. This high impedance state latches the output to low when the previous state is low (Fig. 10(d)). The half-SRAM circuit can also be realized with measured hold static noise margin (HSNM) and read static noise margin (RSNM) (Fig. 11). Improving SRAM performance can be achieved by reducing isolation leakage. The benchmark of transistor stacking is shown in Fig. 12. Our monolithic three-tier transistor stacking is demonstrated with the measured tri-state inverter and half SRAM for the first time.

Conclusion

The monolithic three-tier transistor stacking can further scale the density beyond CFETs. Split gate process can enable novel DTCO design to gain PPA.

Acknowledgements "Center for Advanced Semiconductor Technology Research" under Higher Education Sprout Project by the MOE, NSTC 113-2218-E-002-039-MBK, MOE (NTU-CC-113J893401), Ma-tek, and TSRI, Taiwan.

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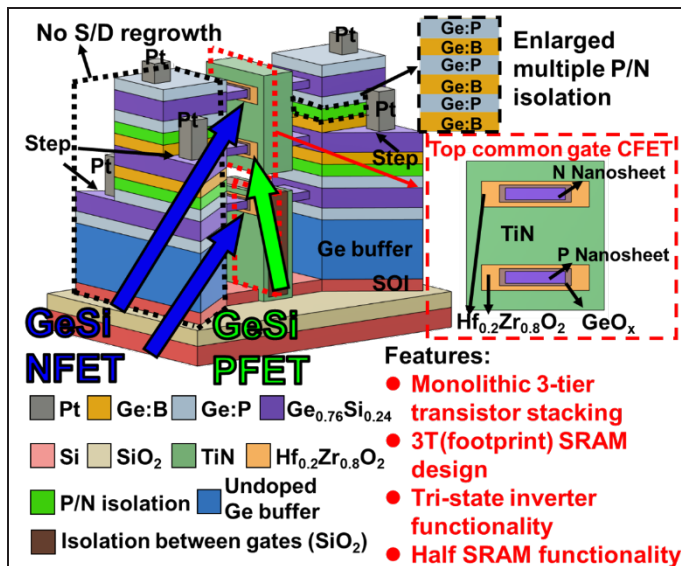


Fig. 1. 3D schematics of three-tier transistor stacking using $\text{Ge}_{0.76}\text{Si}_{0.24}$ nanosheet channels with the highlighted features.

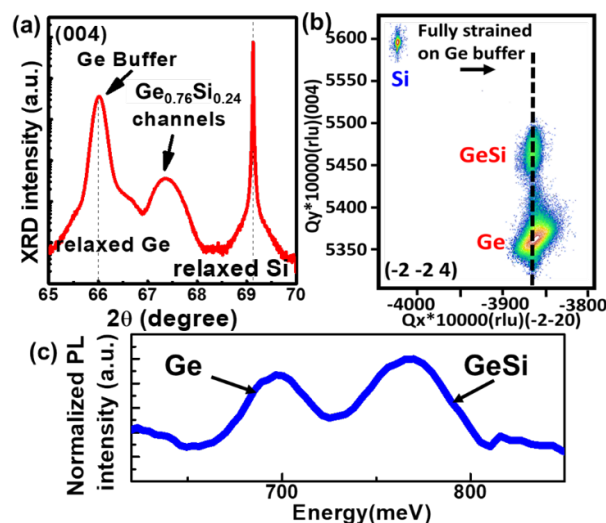


Fig. 3. (a)XRD, (b) RSM and (c) PL of the epilayers. The $\text{Ge}_{0.76}\text{Si}_{0.24}$ channel layers are fully strained on Ge buffer. Strong PL spectra indicates device-quality epi.

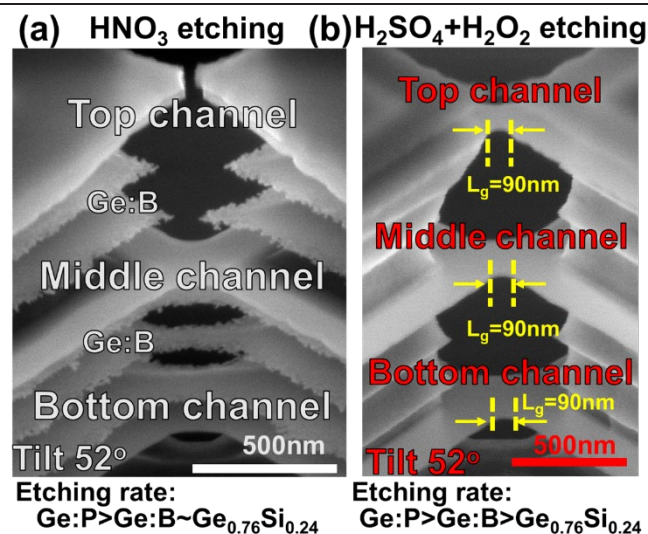


Fig. 5. Tilt 52° SEM image of the three-tier transistor stacking after channel release using (a) HNO_3 (not working) and (b) $\text{H}_2\text{SO}_4+\text{H}_2\text{O}_2$ (working) wet etching.

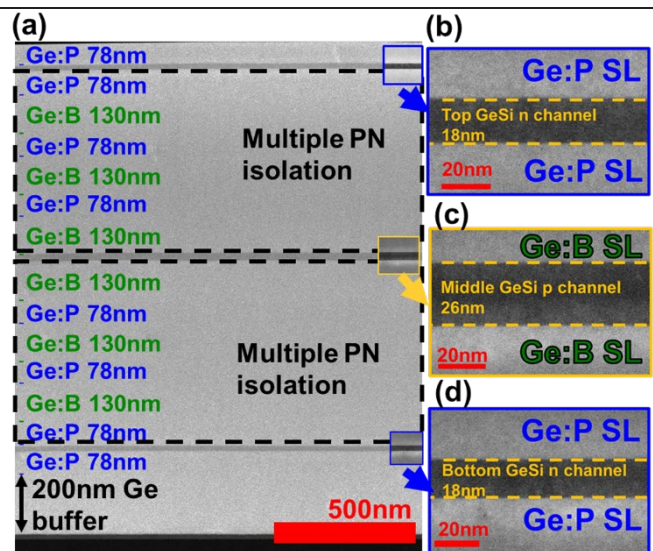


Fig. 2. (a)Epi TEM image and (b-d) three enlarged individual channel epi images. Multiple Ge:B/Ge:P epilayers are grown to electrically isolate the stacked transistors.

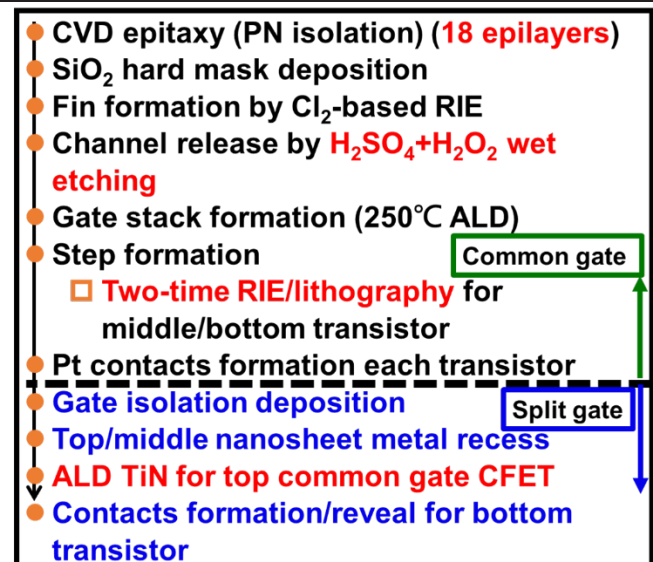


Fig. 4. Key process flow of three-tier NS transistor stacking. The split gate process is after the common gate process.

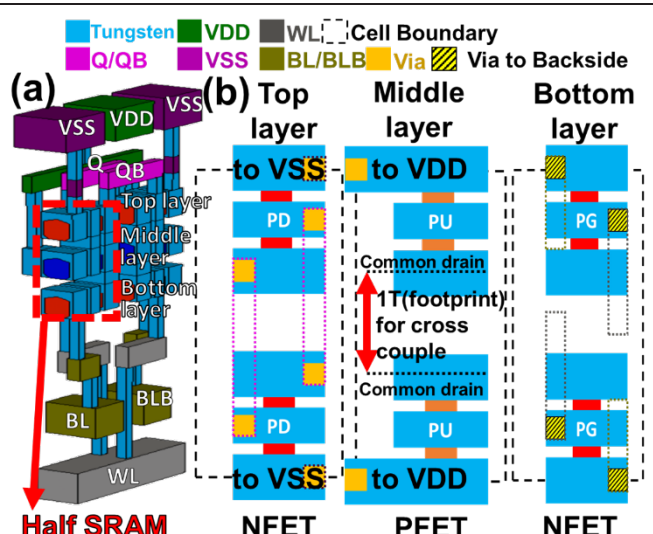


Fig. 6. (a) 3D schematic and (b-d) layout of each layer of three-tier transistor stacking for 3T footprint SRAM design. 1T footprint is used for the cross couple connection between two inverters.

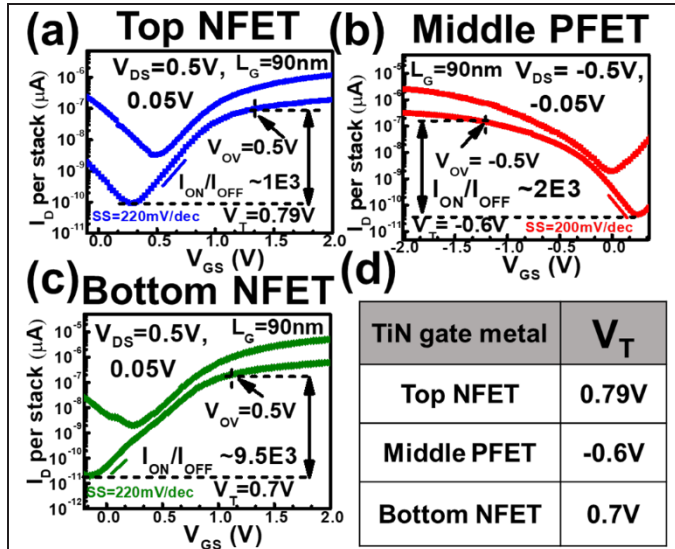


Fig. 7. The transfer characteristics of each transistor in the three-tier transistor stacking with $L_G=90\text{nm}$. The NFETs and PFET can have positive and negative V_T , respectively, due to TiN WFM.

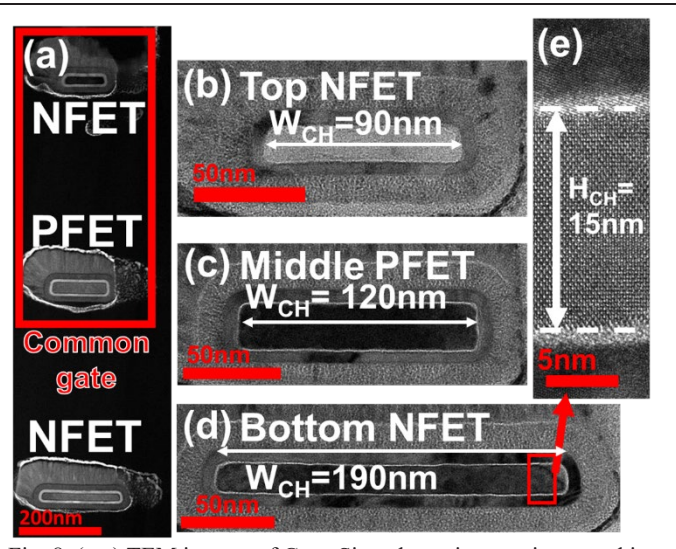


Fig. 8. (a-e) TEM images of $\text{Ge}_{0.75}\text{Si}_{0.25}$ three-tier transistor stacking. The different channel width (W_{CH}) is due to the tapered fin. The fin top has a longer etching time than the bottom.

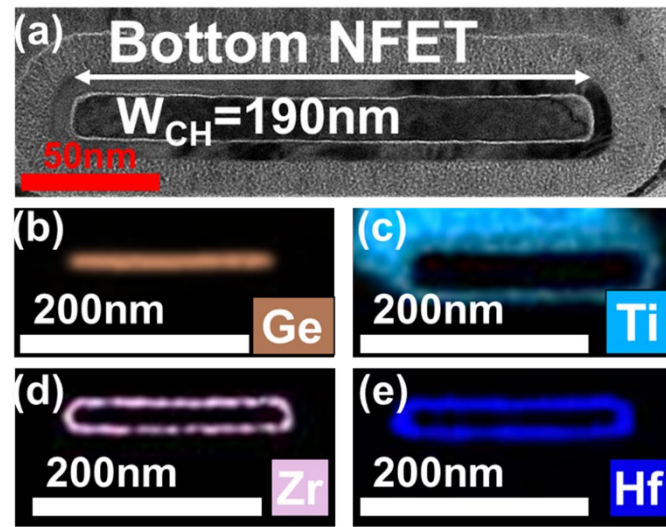


Fig. 9. (a) TEM images and (b-e) EDS mapping of bottom NFET $\text{Ge}_{0.75}\text{Si}_{0.25}$. The EDS mapping confirms that gate stacks wrap around the NS channel. The top and middle FET have similar images.

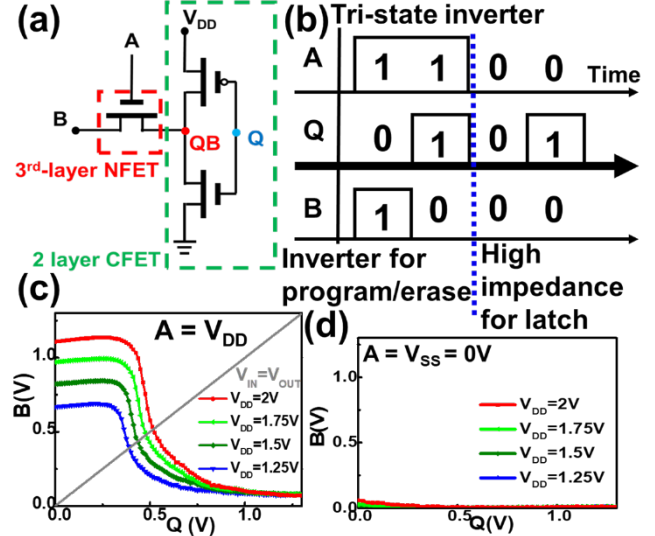


Fig. 10. (a) Tri-state inverter circuit, (b) the I/O waveform, and (c, d) the measured results of the tri-state inverter. The tri-state inverter can be used as an inverter for program/erase or high impedance for latch.

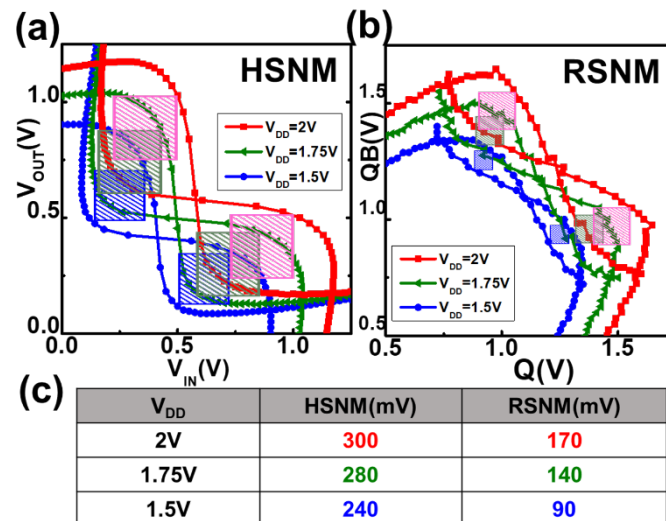


Fig. 11. The measured butterfly curves for (a) HSNM and (b) RSNM of half SRAM using the three-tier transistor stacking.

TABLE I. Benchmark of our three-tier transistor stacking.

Ref.	This work	tsmc[4]	Samsung[11]	TSRI[12]
3D stacked technique	Epitaxy/monolithic	Epitaxy/Monolithic	Laser Crystallization/Sequential	Layer transfer/Sequential
Number of tiers	3 Tier	2 Tier	3 Tier	3 Tier
Channel material	GeSi/GeSi/GeSi	Si/Si	Si/Si/Si	IGZO/Ge/Si
Device	NS	NS	- (possible planar)	Planar/NS/NS
CFET inverter	Yes	Yes	No	Yes
SRAM	Half SRAM (1PD/1PU/1PG)	No	Yes	Simulated
S/D formation	Heavily doped SLs (No S/D regrowth)	Dual S/D epitaxy	-	Ion implantation
Isolation	pn multiple layer	Dielectric	Dielectric	Dielectric